



GISMA University of Applied Sciences

BSc Computer Science

Computer Science Lab

Computer Science Portfolio

Individual Project Report

Aayush Dhami (GH1056547)

June 25, 2026

Contents

1	Introduction	3
2	Portfolio Overview	3
3	Portfolio Website Structure	4
3.1	Home Page	4
3.2	Portfolio Section	5
3.3	Resume Section	5
3.4	About Me Section	6
3.5	Contact Section	6
4	Design Decisions	6
4.1	Colour Scheme	7
4.2	Navigation Bar	7
4.3	Responsive Layout	7
4.4	Project Showcase Layout	7
5	GitHub Repository Structure	7
6	Technologies Used	8
7	Challenges Encountered During Development	9
8	Evaluation and Reflection	9
8.1	Strengths	9
8.2	Weaknesses	10
8.3	Future Improvements	10
9	Conclusion	10
10	References	10

1 Introduction

This report presents the design, development, and evaluation of my personal Computer Science Portfolio website. The portfolio was created as part of the Computer Science Lab module to showcase my technical skills, academic background, projects, and professional experience.

The website was developed using HTML, CSS, and JavaScript and is hosted using GitHub Pages. The portfolio serves as an online professional profile that can be shared with employers and recruiters.

Portfolio Website:

https://aayushdhami578-sketch.github.io/Portfolio_Web/

GitHub Repository:

https://github.com/aayushdhami578-sketch/Portfolio_Web

2 Portfolio Overview

The primary objective of this portfolio website is to present my academic background, technical skills, projects, and professional experience in a structured and professional manner. As a Computer Science student, having an online portfolio allows potential employers, recruiters, and academic staff to evaluate my abilities beyond a traditional curriculum vitae. The portfolio serves as a central location where visitors can learn about my education, view my projects, and access my contact information.

The website is organised into five main sections: Home, Portfolio, Resume, About Me, and Contact. Each section was designed with a specific purpose and contributes to the overall presentation of my professional profile.

The Home section acts as the landing page of the website. It provides a simple greeting, introduces me by name, and presents a profile photograph. The purpose of this section is to create a positive first impression and clearly identify the website as my personal computer science portfolio.

The Portfolio section showcases several academic and personal projects that demonstrate my technical abilities. These include the Student Profile Management System, Invoice Management System, OTAKUwears Website, and this Portfolio Website. Each project is accompanied by a screenshot and a brief description to help visitors understand the purpose and technologies involved.

The Resume section presents my educational qualifications and professional experience. It highlights my studies in Computer Science, foundation year education, and teaching experience as a German Language Instructor. This section allows visitors to quickly review my academic and professional background.

The About Me section provides additional information about my interests, career goals, and technical skills. It includes a personal summary and a collection of skill badges representing technologies and tools that I have experience using, such as HTML, CSS, JavaScript, Python, Git, GitHub, and LaTeX.

The Contact section contains my contact details, including GitHub, LinkedIn, phone number, and location. It also includes a contact form that demonstrates form design and user interface implementation. This section enables visitors to find different ways of communicating with me.

The overall structure was intentionally kept simple and easy to navigate. Rather than creating multiple webpages, a single-page design was selected because it allows users to access all information by scrolling through clearly defined sections. This approach provides a smooth user experience while maintaining a clean and professional presentation.

3 Portfolio Website Structure

The website follows a modern single-page portfolio design.

The design uses a grey colour scheme combined with white text to maintain readability and a professional appearance. Responsive layouts were implemented using CSS Flexbox and Media Queries to ensure compatibility across different screen sizes.

Key design objectives included:

- Simple navigation
- Professional appearance
- Responsive design
- Easy access to projects and CV
- Clear presentation of information

3.1 Home Page

The home page introduces visitors to the portfolio and provides a brief professional summary.

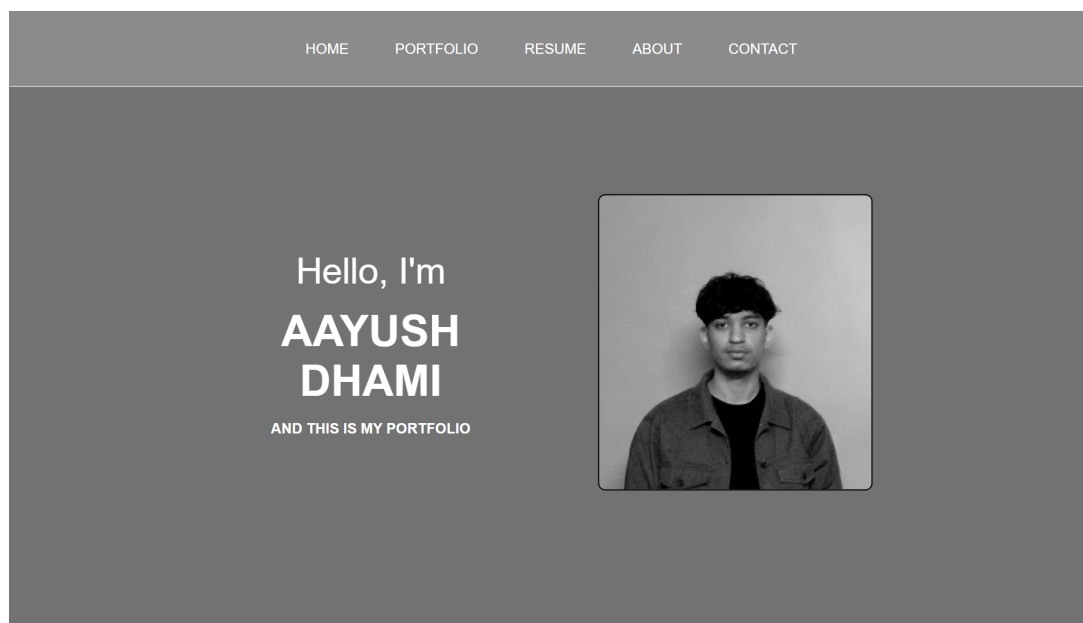


Figure 1: Home Page

3.2 Portfolio Section

This section showcases my technical projects. Each project includes an image, description, technologies used, and links to the source code where appropriate.

Projects included:

- Invoice Management System (C++)
- Student Profile Management System (Python)
- Online Store Website (HTML, CSS, JavaScript)

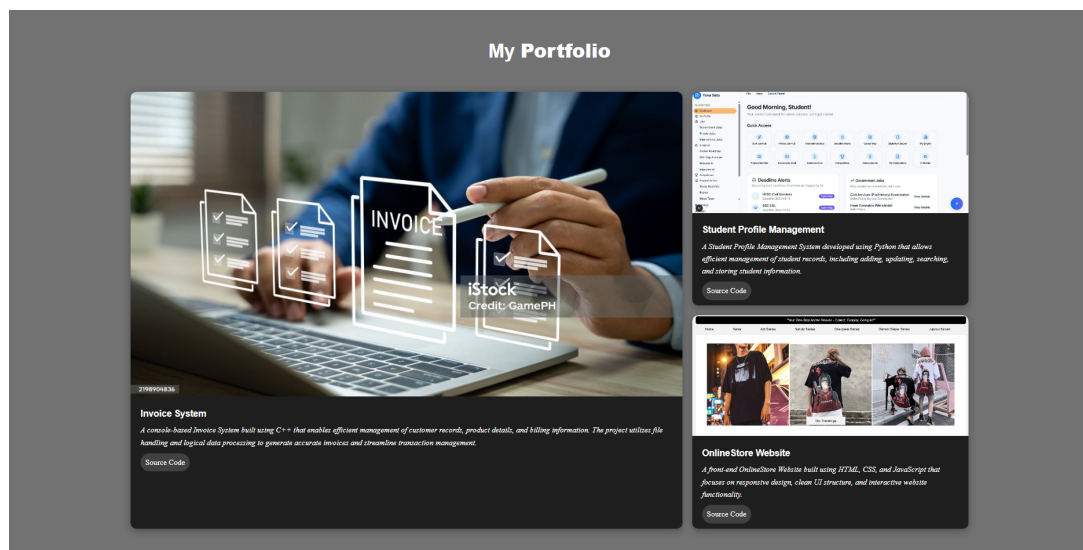


Figure 2: Portfolio Projects

3.3 Resume Section

The Resume section displays my education and work experience.

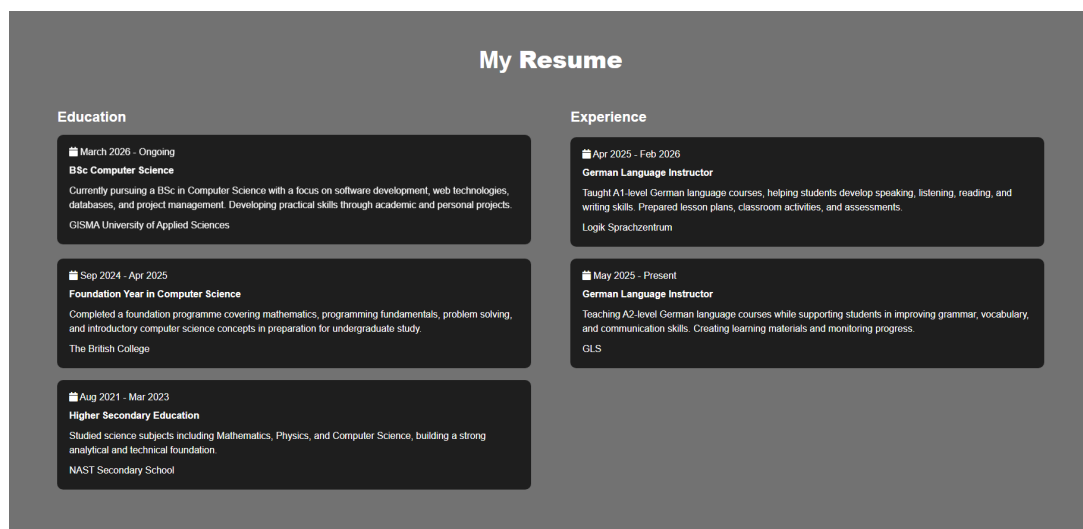


Figure 3: Resume Section

3.4 About Me Section

This section provides information about my background, interests, and technical skills.

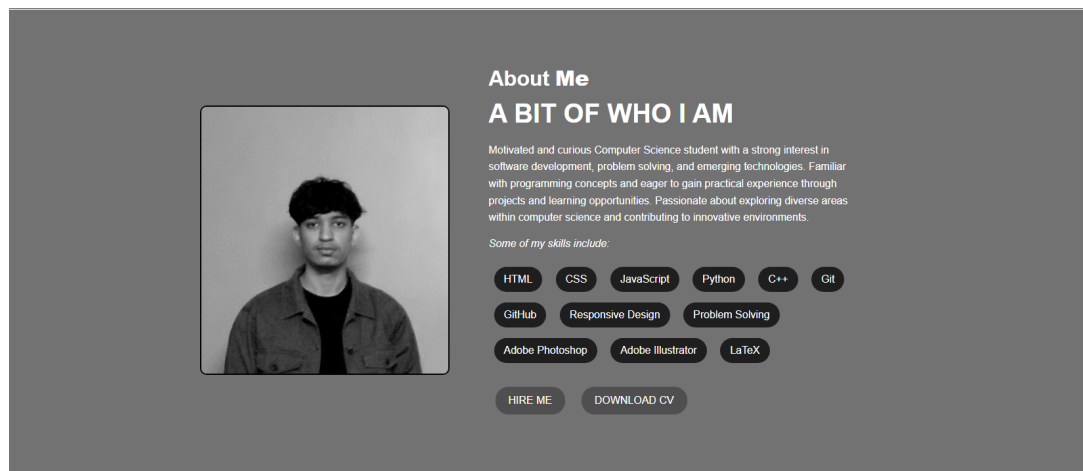


Figure 4: About Me Section

3.5 Contact Section

The contact section allows visitors to access my contact details and submit messages.

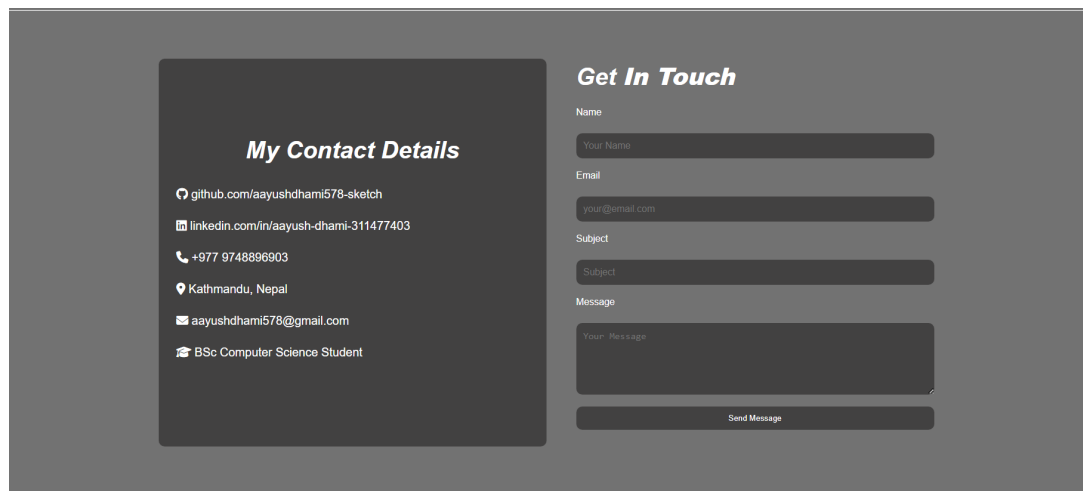


Figure 5: Contact Section

4 Design Decisions

Several design decisions were made throughout the development of the portfolio website to improve usability, readability, responsiveness, and overall visual appeal. The objective was to create a professional portfolio that presents information clearly while remaining accessible across different devices.

4.1 Colour Scheme

A dark grey colour palette was selected because it provides a modern and professional appearance that is commonly found in technology-related websites and portfolios. Dark backgrounds help important content stand out while reducing unnecessary visual distractions. White text was used throughout the website to maximise readability and provide strong contrast against the background. Consistent colours were maintained across all sections to create a unified visual identity and improve the overall user experience.

4.2 Navigation Bar

A sticky navigation bar was implemented to improve website navigation. The navigation menu remains visible while users scroll through the page, allowing quick access to different sections without needing to return to the top of the website. JavaScript was used to highlight the currently active section, providing visual feedback and helping users understand their location within the portfolio. This improves usability, particularly on longer single-page websites.

4.3 Responsive Layout

One of the most important design considerations was ensuring that the website functioned correctly on different screen sizes. During the initial stages of development, some sections relied on fixed positioning and dimensions. While the layout appeared correct on my personal computer, testing revealed that the website became distorted on devices with different screen resolutions.

To address this issue, the layout was redesigned using CSS Flexbox, CSS Grid, and Media Queries. Flexbox was used to organise content into flexible rows and columns, while Grid was used to structure the project showcase section. Media Queries were implemented to adjust layouts automatically on smaller screens. These changes significantly improved responsiveness and ensured that the portfolio remained usable on desktop computers, tablets, and mobile devices.

4.4 Project Showcase Layout

The Portfolio section uses a grid-based layout to display projects in a visually organised manner. A larger project card was used to highlight a featured project, while smaller cards were used for supporting projects. This creates a clear visual hierarchy and helps direct the visitor's attention toward key pieces of work. Each project includes a screenshot and description, allowing visitors to quickly understand the purpose and functionality of the showcased work.

5 GitHub Repository Structure

The project is managed using Git and hosted on GitHub. Version control was used throughout development to track changes and maintain project history.

The repository contains:

- index.html – Main website structure

- style.css – Website styling and layout
- script.js – Interactive website functionality
- Image folder – Project screenshots and profile images
- CV.pdf – Curriculum Vitae created using LaTeX
- README.md – Repository documentation

This structure ensures that content, styling, and functionality remain separated and easy to manage.

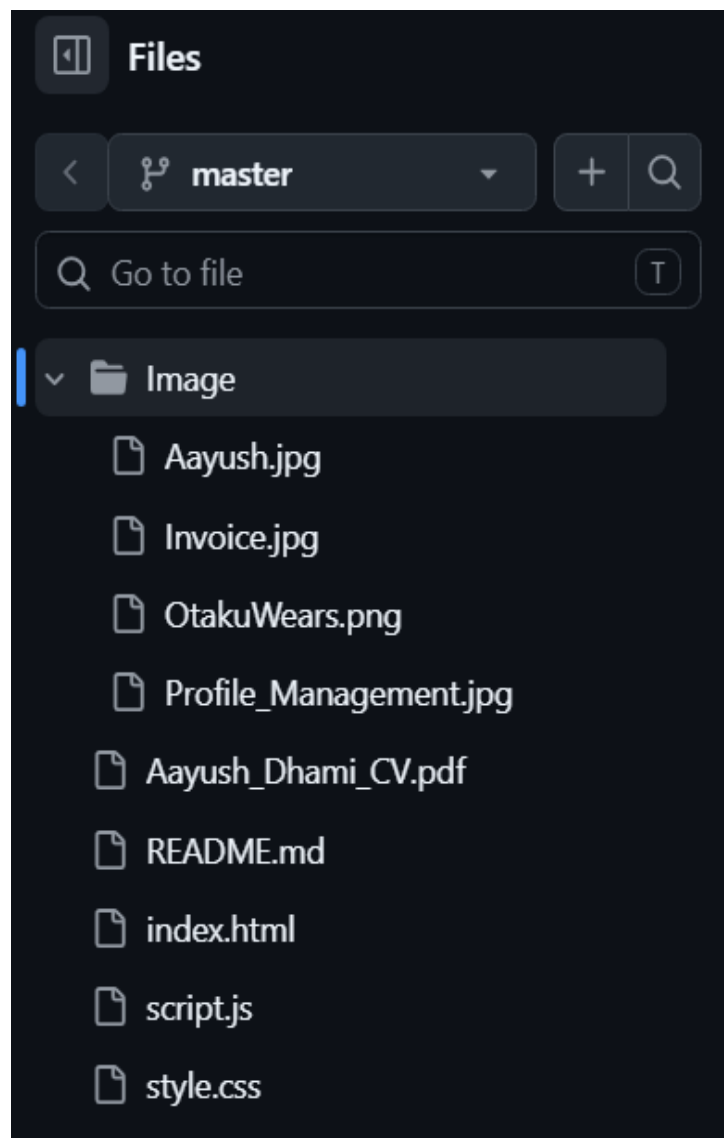


Figure 6: GitHub Repository Structure

6 Technologies Used

Several technologies were used during development:

- **HTML:** HTML was used to create the structure of the website, including sections, navigation, project cards, and content.
- **CSS:** CSS was used to style the website and create layouts using Flexbox and Grid. It was also used to implement animations, hover effects, and responsive design.
- **JavaScript:** JavaScript was used to improve user interaction, including the active navigation highlighting feature and other dynamic elements.
- **Git and GitHub:** Git was used for version control, while GitHub was used to host the repository and publish the portfolio through GitHub Pages.
- **LaTeX :** LaTeX was used to create a professional Curriculum Vitae. The CV was included within the repository and linked from the portfolio website.
- **Visual Studio Code :** Visual Studio Code was used as the development environment.

These projects demonstrate skills in programming, web development, version control, and software design.

7 Challenges Encountered During Development

Several challenges were encountered throughout the development process. One of the first challenges involved deploying the website using GitHub Pages. Although the website functioned correctly in the local development environment, certain images failed to appear after deployment. After investigating the issue, it was discovered that file path names were not matching the exact capitalisation of the folder names stored in the repository. Because GitHub Pages operates on a Linux-based system, file paths are case-sensitive. Correcting the folder names resolved the problem.

Another challenge involved creating a responsive layout. Initially, many elements relied on fixed dimensions and absolute positioning. This approach produced acceptable results on a single device but created problems when viewed on different screen sizes. Through testing and experimentation, the layout was restructured using Flexbox and Media Queries. This significantly improved responsiveness and reduced layout inconsistencies.

Version control also presented a learning opportunity. Throughout development, Git and GitHub were used to manage source code and track changes. Working with commits, branches, and repository updates improved my understanding of software development workflows and highlighted the importance of maintaining organised project histories.

8 Evaluation and Reflection

8.1 Strengths

One of the strongest aspects of the portfolio is its organisation. Information is grouped logically into clearly defined sections, allowing visitors to navigate efficiently. The use of GitHub Pages provides a reliable and publicly accessible hosting platform, while responsive design techniques ensure compatibility across different devices.

The project also demonstrates practical knowledge of HTML, CSS, JavaScript, Git, GitHub, and LaTeX. The integration of these technologies highlights both technical and organisational skills that are valuable within software development environments.

8.2 Weaknesses

Despite these strengths, several limitations remain. The contact form currently lacks backend functionality and therefore cannot process user submissions. Additionally, JavaScript functionality is relatively limited and could be expanded to create a more interactive user experience. The portfolio would also benefit from the inclusion of additional projects that demonstrate more advanced programming concepts.

8.3 Future Improvements

Although the portfolio successfully achieves its primary objectives, several improvements could be implemented in future versions. One potential enhancement would be the addition of more software development projects. Including larger and more technically advanced projects would provide visitors with a broader understanding of my programming abilities and problem-solving skills.

Another important improvement would be the implementation of a fully functional backend for the contact form. At present, the form serves primarily as a visual component and does not process user submissions. Integrating a backend service would allow visitors to send messages directly through the website.

Additional interactive features could also improve the user experience. Examples include project filtering, animations, theme customisation, and a dark/light mode switch. These features would increase engagement while demonstrating additional front-end development skills.

Finally, the website could be expanded with a dedicated blog section where technical learning experiences, project updates, and programming insights could be shared. This would allow the portfolio to evolve beyond a static website and become a platform for continuous professional development.

9 Conclusion

This project successfully demonstrates my ability to design, develop, and deploy a professional computer science portfolio website. The portfolio showcases my educational background, technical skills, and projects while providing an online presence that can be shared with employers and recruiters.

The project also allowed me to gain practical experience with web development, Git version control, GitHub Pages deployment, and technical documentation using LaTeX.

10 References

- Overleaf (2026) *Clean CV Template*. Available at: <https://www.overleaf.com/latex/templates/clean-cv-template/rcqjkxnfsmnz> (Accessed: 25 June 2026).

- ThemeWagon (2026) *Rezume Portfolio Template*. Available at: <https://themewagon.github.io/rezume/> (Accessed: 25 June 2026).